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Experiment: 6:5G NR Frame Structure Analysis with Different Subcarrier Spacing

.

Objective – 1: Analyze Different Subcarrier Spacing Values

MATLAB Program

```
clc;
clear;
close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```

Expected Output

• Bar graph showing:

o 15 kHz

o 30 kHz

o 60 kHz

o 120 kHz

o 240 kHz

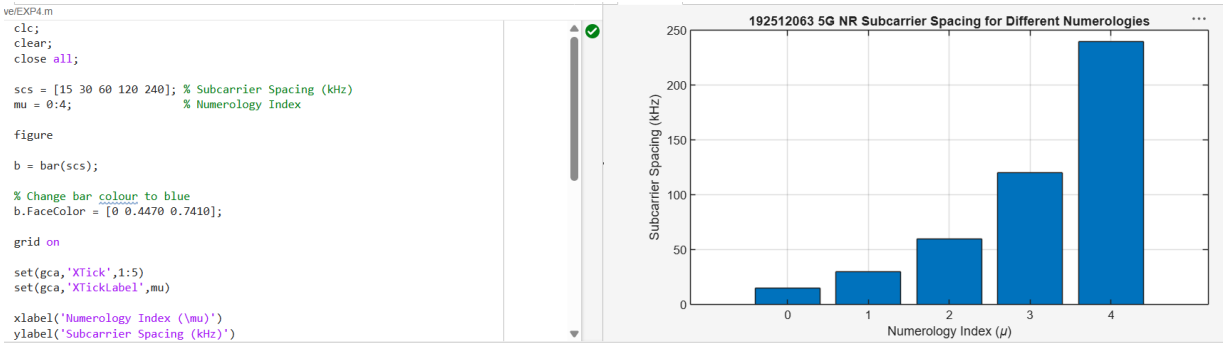


FIG.1.

MATLAB Program

```
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)
plot(scs, slot, 'o-', 'LineWidth', 2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
```

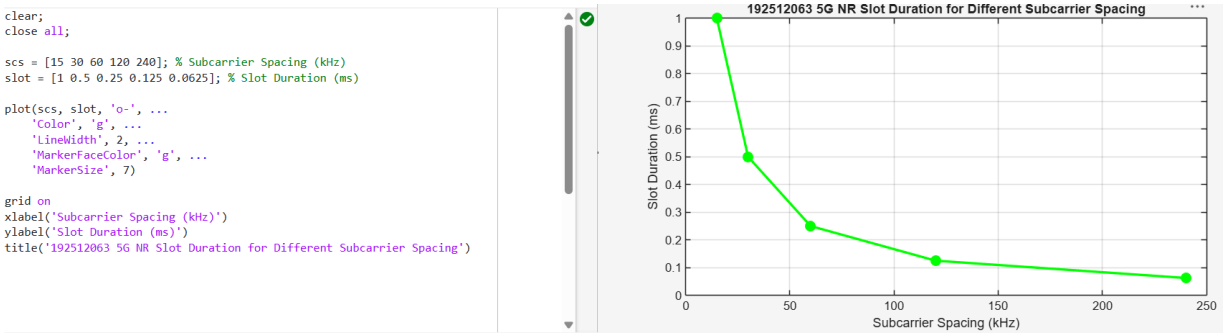


FIG.2.

Expected Output

• Graph showing slot duration decreasing as subcarrier spacing increases.

Objective – 3 Study the Number of OFDM Symbols

MATLAB Program

clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

symbols = [14 14 14 14 14]; % OFDM Symbols per Slot

figure

bar(scs, symbols)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Number of OFDM Symbols')

title('OFDM Symbols per Slot for Different Subcarrier Spacing')

ylim([0 16])

Expected Output

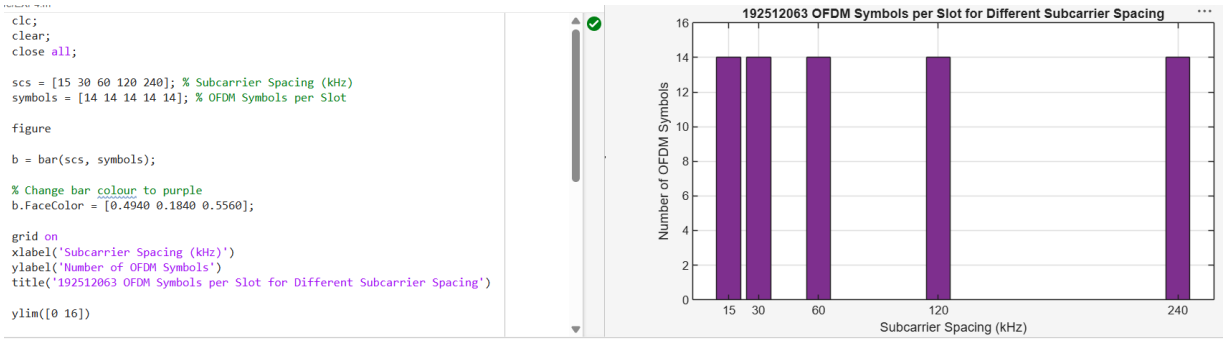


FIG.3.